

Mex Grid Queries

Time limit: 1.00 s

Memory limit: 512 MB

Consider a two-dimensional grid whose rows and columns are 1-indexed. Each square contains the smallest nonnegative integer that does not appear to the left on the same row or above on the same column.

Your task is to calculate the value at square (y, x) .

Input

The only input line contains two integers y and x .

Output

Print one integer: the value at square (y, x) .

Constraints

- $1 \leq y, x \leq 10^9$

Example

Input	Output
3 5	6